Day 089 | Unit 06 | Teacher plan

### Understand and Use Ratios and Rates

# Apply Ratio and Rate Reasoning

Standards: 6.RP.A.3a, 6.RP.A.3b, 6.RP.A.3d

Learning target: I can apply Ratio and Rate Reasoning, and explain and check my reasoning.

enVision Topic 5: Understand and Use Ratios and Rates | Spectrum companion Chapter 3

## 0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

## 5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating. Divide total cost by the number of items or units. Compare costs for exactly one of the same unit. A lower package price does not always give a lower unit price. Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

## 17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

## 42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

## Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

## Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 2

Day 089 | Unit 06 | Lesson

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## 01 / Learn

Multiply both values in a ratio by the same scale factor. Organize pairs in a table and check that every row compares the same quantities in the same order. At a constant speed, distance = speed × time. Divide distance by time to find speed, or distance by speed to find time. Keep units consistent before calculating. Divide total cost by the number of items or units. Compare costs for exactly one of the same unit. A lower package price does not always give a lower unit price. Write an equality such as 1 foot = 12 inches. Multiply to change larger units into smaller units; divide for the reverse. A conversion factor is a ratio equal to 1, so the amount stays the same.

## Worked example 1

A mix uses 7 cups of oats for 2 cups of seeds. Complete the table (oats, seeds): (7, 2), (14, ___), (35, ___).

1. Multiply both values in a ratio by the same scale factor.

2. Organize pairs in a table and check that every row compares the same quantities in the same order.

3. Multiply both parts by 2 and by 5.

4. State the result in the form requested: 4; 10.

Answer: 4; 10

## Worked example 2

Use 1 feet = 12 inches. Select all statements equivalent to 5/2 feet.
A. 31 inches
B. 5/24 inches
C. 30 inches
D. 5/2 × 12 inches

1. Write an equality such as 1 foot = 12 inches.

2. There are 12 smaller units in each larger one.

3. Multiply 5/2 by 12, giving 30 inches.

4. The physical length does not change.

Answer: C, D - 30 inches; 5/2 × 12 inches

# Day 089 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

### 01 / Constructed response

A mix uses 6 cups of oats for 5 cups of seeds. Complete the table (oats, seeds): (6, 5), (12, ___), (18, ___).

### 02 / Constructed response

A bus travels at 25 km/h. How long does 175 km take?

### 03 / Constructed response

Pack A has 2 markers for $12; Pack B has 7 markers for $45.5. Find each unit price and the better buy.

### 04 / Constructed response

Convert 120 minutes to hours. Show the conversion ratio.

### 05 / Constructed response

A mix uses 7 cups of oats for 5 cups of seeds. Complete the table (oats, seeds): (7, 5), (14, ___), (21, ___).

### 06 / Constructed response

A bus travels at 40 km/h. How long does 160 km take?

## 07 / Constructed response

Pack A has 2 markers for $7; Pack B has 7 markers for $28. Find each unit price and the better buy. How much would 14 markers cost using each pack size?

## 08 / Multiple select

Use 1 feet = 12 inches. Select all statements equivalent to 9/2 feet.
A. 54 inches
B. 3/8 inches
C. 55 inches
D. 9/2 × 12 inches

## 09 / Multiple select

A recipe uses 7 cups oats for 4 cups seeds. Select all (oats, seeds) pairs that keep this ratio.
A. (14, 9)
B. (14, 8)
C. (36, 20)
D. (21, 12)

## 10 / Constructed response

A bus travels at 20 km/h. How long does 60 km take?

Day 089 | Unit 06 | Answer key

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## 1. 10; 15

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

## 2. 7 hours

Time = distance ÷ speed = 175 ÷ 25 = 7.

6.RP.A.3b | Constructed response

## 3. A $6/marker; B $6.5/marker; A is cheaper

Divide each price by its marker count; compare 6 with 6.5.

6.RP.A.3b | Constructed response

## 4. 2 hours

120 ÷ 60 = 2; 1 hours = 60 minutes.

6.RP.A.3d | Constructed response

## 5. 10; 15

Multiply both parts by 2 and by 3.

6.RP.A.3a | Constructed response

## 6. 4 hours

Time = distance ÷ speed = 160 ÷ 40 = 4.

6.RP.A.3b | Constructed response

## 7. A $3.5/marker; B $4/marker; A is cheaper; A $49, B $56

Divide each price by its marker count; compare 3.5 with 4.

6.RP.A.3b | Constructed response

## 8. A, D - 54 inches; 9/2 × 12 inches

There are 12 smaller units in each larger one. Multiply 9/2 by 12, giving 54 inches. The physical length does not change.

6.RP.A.3d | Multiple select

## 9. B, D - (14, 8); (21, 12)

Equivalent ratios multiply both quantities by the same factor. A uses 2; B uses 3. C and D change only one scaled quantity.

6.RP.A.3a | Multiple select

## 10. 3 hours

Time = distance ÷ speed = 60 ÷ 20 = 3.

6.RP.A.3b | Constructed response